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The Log of Gravity

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estimator based on the Poisson distribution is optimal in this situation. Case 3 is the special case in which OLS estimation of the log-linear model is consistent for the slope parameters of (14). Moreover, in this case the log-linear model not only corrects the heteroskedasticity in the data, but, because η_i is log-normal, it is also the maximum likelihood estimator. The GPML is the optimal PML estimator in this case, but it should be outperformed by the true maximum likelihood estimator. Finally, Case 4 is the only one in which the conditional variance does not depend exclusively on the mean. The variance is a quadratic function of the mean, as in Case 3, but it is not proportional to the square of the mean.

We carried out two sets of experiments. The first set was aimed at studying the performance of the estimators of the multiplicative and the log-linear models under different patterns of heteroskedasticity. In order to study the effect of the truncation on the performance of the OLS, and given that this data generating mechanism does not produce observations with $y_i = 0$, the log-linear model was also estimated using only the observations for which $y_i > 0.5$, OLS ($y > 0.5$). This reduces the sample size by about 25% to 35%, depending on the pattern of heteroskedasticity. The estimation of the threshold Tobit was also performed using this dependent variable. Notice that, although the dependent variable has to cross a threshold to be observable, the truncation mechanism used here is not equal to the one assumed by Eaton and Tamura (1994). Therefore, in all these experiments the ET-Tobit will be slightly misspecified and the results presented here should be viewed as a check of its robustness to this problem.

The second set of experiments studied the estimators' performance in the presence of rounding errors in the dependent variable. For that purpose, a new random variable was generated rounding to the nearest integer the values of y_i obtained in the first set of simulations. This procedure mimics the rounding errors in official statistics and generates a large number of zeros, a typical feature of trade data. Because the model considered here generates a large proportion of observations close to zero, rounding down is much more frequent than rounding up. As the probability of rounding up or down depends on the covariates, this procedure will necessarily bias the estimates, as discussed before. The

We argue that the biases are present both in the traditional specification of the gravity equation and in the Anderson-van Wincoop (2003) specification, which includes country-specific fixed effects.

To address the various estimation problems, we propose a simple Poisson pseudo-maximum likelihood method and assess its performance using Monte Carlo simulations. We find that in the presence of heteroskedasticity the standard methods can severely bias the estimated coefficients, casting doubt on previous empirical findings. Our method, instead, is robust to different patterns of heteroskedasticity and, in addition, provides a natural way to deal with zeroes in trade data.

We use our method to re-estimate the gravity equation and document significant differences from the results obtained using the log-linear method. For example, income elasticities in the traditional gravity equation are systematically smaller than those obtained with log-linearized OLS regressions. In addition, in both the traditional and Anderson-van Wincoop specifications of the gravity equation, OLS estimation exaggerates the role of geographical proximity and colonial ties. OLS estimates of the impact of preferential-trade agreements are very sensitive to the inclusion of fixed effects, whereas Poisson generates more stable results. RESET tests systematically favour the Poisson PML technique. The results suggest that heteroskedasticity (rather than truncation of the data) is responsible for the main differences.

Log-linearized equations estimated by OLS are of course used in many other areas of empirical economics and econometrics. Our Monte Carlo simulations and the regression outcomes indicate that in the presence of heteroskedasticity this practice can lead to significant biases. These results suggest that, at least when there is evidence of heteroskedasticity, the Poisson pseudo-maximum likelihood estimator should be used as a substitute for the standard log-linear model.